| Component | Part Number | Quantity | Unit Price |
|---|---|---|---|
| Buck IC | AP64200SP-13 | 1 | $1.38 |
| Inductor | SRR1210A-100M | 1 | $1.47 |
| Input capacitor | C1210C106K5RACTU, 10 µF, 50 V | 2 | $1.88 |
| Output capacitor | C1210C226M4RACTU, 22 µF, 16 V | 2 | $2.81 |
| Bootstrap capacitor | CC0603KRX7R9BB104, 100 nF | 1 | $0.14 |
| Feedback resistor R1 | RC0603FR-0752K3L, 52.3 kΩ | 1 | $0.10 |
| Feedback resistor R2 | RC0603FR-0710KL, 10 kΩ | 1 | $0.10 |
| Frequency resistor RT | CR0603-FX-2003ELF, 200 kΩ | 1 | $0.10 |
| Compensation resistor R5 | RC0603FR-0714KL, 14 kΩ | 1 | $0.11 |
| Compensation capacitor C5 | CC0603KRX7R9BB562, 5.6 nF | 1 | $0.12 |

| Component | DigiKey Link |
|---|---|
| Buck IC | AP64200SP-13 |
| Inductor | SRR1210A-100M |
| Input capacitor | 10 µF capacitor |
| Output capacitor | 22 µF capacitor |
| Bootstrap capacitor | 100 nF capacitor |
| Feedback resistor R1 | 52.3 kΩ resistor |
| Feedback resistor R2 | 10 kΩ resistor |
| Frequency resistor RT | 200 kΩ resistor |
| Compensation resistor R5 | 14 kΩ resistor |
| Compensation capacitor C5 | 5.6 nF capacitor |

Buck IC:
- I selected the AP64200 because it supports the required input voltage and can provide up to 2A

Inductor:
- I selected a 10uH inductor because it helps reduce current ripple. It also has a 6.5 A current rating, which is sufficient for the design.

Input Capacitors:
- I selected two 10uF capacitors to reduce input voltage ripple and switching noise

Output Capacitors:
- I selected two 22uF capacitors to smooth the output voltage

Bootstrap Capacitor:
- The 100nF capacitor helps operate the high-side switching transistor

Feedback Resistors:
- The 52.3kohm and 10kohm resistors set the output voltage to approximately 5V

Frequency Resistor:
- The 200kohm resistor sets the switching frequency to approximately 500kHz

Compensation Components:
- The 14kohm resistor and 5.6 nF capacitor help stabilize the converter